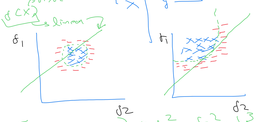
↳ ML algo → Intro → devise
↳ supervised → target



[feature eng] → θ_1^2 θ_2^2 $\theta_1\theta_2$ θ_1^3 θ_2^3 ...

[polynomial features OVR / OVA]

sin
cos
($\theta_1\theta_2$)

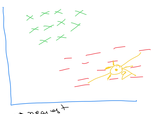
θ_1 θ_2 θ_3 → multi-collinear

$\theta_1 = \alpha\theta_2 + \beta\theta_3 + \gamma$

KNN

A ↔ B (similar)

Assumption: K N N → neighbours → will have similar traits (homogeneous)



KNN → neighbours → hyper-parameters → mean of t

Data → Train → D_T
Data → Test → hyper-parameter tuning

x_q

① x_q → D_T (n=1000) euclidean dist

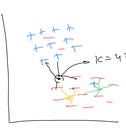
$x_{21} \to x_q$ $d_1 \to 2.5$
$x_{22} \to x_q$ $d_2 \to 1.9$
$x_{23} \to x_q$ $d_3 \to 3.8$
⋮
$x_{1000} \to x_q$ $d_{1000} \to 15.7$

$\vec{x_i, y_i}$ $\vec{x_j, y_j}$ → (0,0)
$(x_i - x_j)^2 + (y_i - y_j)^2$

② sort → ascending order.

X	d	Y
x_{21}		
x_{22}		
x_{23}		
x_{1000}		

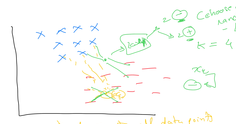
k top nearest neighbours



⊖ → not even
⊙ → multiple (5)

k = 5

5 — ⊖ → majority
+ ⊙
5 — ⊖ → majority −
+ ⊙

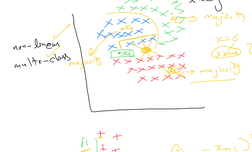


z ⊙ (choose at random)

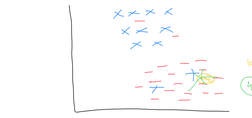
① find dist of x_q to all data points in training data
② sort them in ascending order
③ pick top k data points
④ find the majority class in training data points

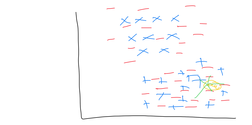
k → 3

W → non parameter algorithm / lazy learner



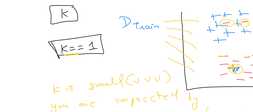
$(x_{q1} - x_{n1})^2$
$(x_{q2} - x_{n2})^2$
(0,1)





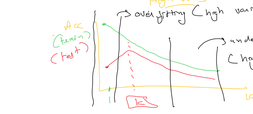
k

k==1



k is small (a vvv) → you are impacted by outliers → high variance | overfit

n==4 k==81 prediction → mode
high bias



Prediction
→ euclidean dist → n → d
↳ n×d
↳ sort → n log n
↳ top k → k

k + nd + n log n
nd + n log n

n×d → space

$n(d + \log n)$ → speed

training O(1)